

SSS 1-3 Sensor and Monitoring Garden Flashcards 921-940

921. MEASUREMENT SYSTEM	A measurement system includes the measurand, sensor, signal conditioning, data a...
922. SENSOR SPECIFICATION	Sensor specifications define range, accuracy, resolution, response, environmen...
923. ACCURACY	Accuracy describes closeness to a reference or true value under specified condit...
924. PRECISION	Precision describes agreement among repeated measurements and can be quantified ...
925. BIAS ERROR	Bias is a systematic measurement error that shifts readings consistently away fr...
926. RANDOM ERROR	Random error produces unpredictable measurement variation and contributes to unc...
927. CALIBRATION CURVE	A calibration curve relates sensor output to known reference values and may be u...
928. VALIDATION DATA	Validation uses independent reference observations to test sensor performance af...
929. RMSE	Root mean square error summarizes the magnitude of prediction or measurement err...
930. MAE	Mean absolute error is the average absolute difference between measured or predi...
931. SAMPLING FREQUENCY	Sampling frequency is how often a signal is measured and must be appropriate for...
932. DATA LOGGER	A data logger time-stamps, stores, and sometimes transmits measurements from one...
933. METADATA	Metadata document sensor identity, units, location, depth, calibration, timestam...
934. TIME-SERIES QUALITY CONTROL	Time-series quality control detects missing values, impossible values, spikes, d...
935. OUTLIER DIAGNOSIS	An outlier should be investigated using context, neighboring measurements, instr...
936. MISSING-DATA PATTERN	Missing-data patterns can reveal power, communication, logger, sensor, or mainte...
937. SENSOR PLACEMENT	Placement determines whether measurements represent the intended spatial locatio...
938. DATA INTEGRATION	Data integration combines measurements from multiple variables, sensors, locatio...
939. MONITORING DASHBOARD	A monitoring dashboard summarizes current state, trends, thresholds, data qual...
940. MONITORING SYSTEM EVALUATION	Evaluation considers sensor fitness, calibration, uncertainty, placement, data c...